Behavioral Inequality: The Contribution of Decision-Making Frictions to Inequality

Stefano DellaVigna
UC Berkeley & NBER

Tim de Silva
Stanford GSB

Rohan Jha
Harvard

August 2026

Preliminary and Incomplete ☺

What We Do

- Individuals make many **choices** that are hard to justify with neoclassical models
  - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

What We Do

- Individuals make many **choices** that are hard to justify with neoclassical models
  - Suboptimal refinancing, not claiming benefits, dominated insurance, ...
- Our questions:
  1. What are the **financial impacts** of these “frictions”, individually and in aggregate?
  2. How much do they contribute to **inequality** in income, consumption, and wealth?

What We Do

- Individuals make many **choices** that are hard to justify with neoclassical models
  - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

- Our questions:
  1. What are the **financial impacts** of these “frictions”, individually and in aggregate?
  2. How much do they contribute to **inequality** in income, consumption, and wealth?

- Approach #1: **Empirical framework** + existing literature + existing data

  $\Rightarrow$ Estimates of losses across income distribution that are comparable across frictions

What We Do

- Individuals make many **choices** that are hard to justify with neoclassical models
  - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

- Our questions:
  1. What are the **financial impacts** of these "frictions", individually and in aggregate?
  2. How much do they contribute to **inequality** in income, consumption, and wealth?

- Approach #1: **Empirical framework** + existing literature + existing data

  ⇒ Estimates of losses across income distribution that are comparable across frictions

- Approach #2: Combine Approach #1 with a **life cycle model**

  ✓ Accounts for dynamic effects and endogenous responses

  ⇒ Estimates of effects on aggregate consumption and wealth inequality

What We Do

- Individuals make many **choices** that are hard to justify with neoclassical models
  - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

- Our questions:
  1. What are the **financial impacts** of these “frictions”, individually and in aggregate?
  2. How much do they contribute to **inequality** in income, consumption, and wealth?

- Approach #1: **Empirical framework** + existing literature + existing data

  $\Rightarrow$ Estimates of losses across income distribution that are comparable across frictions

- Approach #2: Combine Approach #1 with a **life cycle model**

  ✓ Accounts for dynamic effects and endogenous responses

  $\Rightarrow$ Estimates of effects on aggregate consumption and wealth inequality

Takeaway: Effects of frictions are large, heterogeneous, and regressive

What Do We Mean by a "Friction"?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

What Do We Mean by a "Friction"?

Our Definition of a Friction

A **friction** is a <u>choice</u> made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes <u>normative utility</u> with <u>full information</u> and <u>no non-monetary decision costs</u>.

1. A friction is **not** necessarily a mistake (“improvability”, Bernheim et al. 26)

What Do We Mean by a "Friction"?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

1. A friction is **not** necessarily a mistake (“improvability”, Bernheim et al. 26)

2. We **do not** specify the underlying microfoundation(s)

Scope: Estimate the **financial impact** of a large set of frictions, not

- Evaluate specific policies
- Measure the effects of specific biases
- Assess the welfare implications of removing frictions

What Do We Mean by a "Friction"?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

1. A friction is **not** necessarily a mistake (“improvability”, Bernheim et al. 26)
2. We **do not** specify the underlying microfoundation(s)

   Scope: Estimate the **financial impact** of a large set of frictions, not
   - Evaluate specific policies
   - Measure the effects of specific biases
   - Assess the welfare implications of removing frictions
3. If choice is not a friction, estimates are lower bound on utility costs to rationalize it

List of Frictions That We Consider

No.	Category	Friction	Syllabi Covering Friction		Coverage in Top Journals
			Top Universities (Out of 10)	High Schools (Out of 10)	
1	Education	Not Attending College Due to Frictions	0	10	.
2	Education	Attending 2-Year For-Profit Colleges	0	1	.
3	Savings	Not Meeting the Employer 401(k) Match	3	3	.
4	Savings	Not Investing in Stock Mutual Funds	10	10	.
5	Savings	Saving in Low Interest Accounts	5	4	.
6	Savings	Investing in Individual Stocks	6	7	.
7	Savings	Investing in High-Fee Mutual Funds	4	1	.
8	Borrowing	Not Refinancing a Mortgage	6	2	.
9	Borrowing	Not Getting an Additional Mortgage Quote	1	0	.
10	Borrowing	Suboptimally Repaying Credit Card Debt	0	0	.
11	Borrowing	Paying Late Fees	2	8	.
12	Borrowing	Paying Overdraft Fees	4	6	.
13	Insurance	Choosing Dominated Health Insurance	0	2	.
14	Insurance	Suboptimally Lapsing Life Insurance	1	0	.
15	Benefits	Not Claiming EITC Benefits	1	1	.
16	Benefits	Not Claiming SNAP Benefits	0	0	.
17	Health	Not Taking Statin Medications	0	0	.
18	Health	Not Screening for Colon Cancer	0	0	.
19	Health	Not Quitting Smoking	0	0	.

timdesilva.me/frictions

List of Frictions That We Consider

No.	Category	Friction	Syllabi Covering Friction: Top Universities (Out of 10)	Syllabi Covering Friction: High Schools (Out of 10)	Coverage in Top Journals
1	Education	Not Attending College Due to Frictions	0	10	.
2	Education	Attending 2-Year For-Profit Colleges	0	1	.
3	Savings	Not Meeting the Employer 401(k) Match	3	3	.
4	Savings	Not Investing in Stock Mutual Funds	10	10	.
5	Savings	Saving in Low Interest Accounts	5	4	.
6	Savings	Investing in Individual Stocks	6	7	.
7	Savings	Investing in High-Fee Mutual Funds	4	1	.
8	Borrowing	Not Refinancing a Mortgage	6	2	.
9	Borrowing	Not Getting an Additional Mortgage Quote	1	0	.
10	Borrowing	Suboptimally Repaying Credit Card Debt	0	0	.
11	Borrowing	Paying Late Fees	2	8	.
12	Borrowing	Paying Overdraft Fees	4	6	.
13	Insurance	Choosing Dominated Health Insurance	0	2	.
14	Insurance	Suboptimally Lapsing Life Insurance	1	0	.
15	Benefits	Not Claiming EITC Benefits	1	1	.
16	Benefits	Not Claiming SNAP Benefits	0	0	.
17	Health	Not Taking Statin Medications	0	0	.
18	Health	Not Screening for Colon Cancer	0	0	.
19	Health	Not Quitting Smoking	0	0	.

Outline

1. Empirical Framework and Examples (Approach #1)
2. Results (Approach #1)
3. Life Cycle Model (Sketch) and Results (Approach #2)
4. Conclusion

Outline

1. Empirical Framework and Examples (Approach #1)
2. Results (Approach #1)
3. Life Cycle Model (Sketch) and Results (Approach #2)
4. Conclusion

Environment

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low}, \text{Middle}, \text{High}\}$
  - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}\,(i$ at risk of friction at $t)$, $f_{it} \equiv \mathbf{1}\,(i$ subject to friction at $t)$
  - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
  - f_{it} may depend on past choices (e.g. for-profit college attendance)

Environment

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low}, \text{Middle}, \text{High}\}$
  - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}\,(i \text{ at risk of friction at } t)$, $f_{it} \equiv \mathbf{1}\,(i \text{ subject to friction at } t)$
  - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
  - f_{it} may depend on past choices (e.g. for-profit college attendance)
- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \underbrace{\overbrace{\text{Wealth}_{it+1}(f_{it})}^{\text{carried out of } t} - \overbrace{\text{Wealth}_{it}(f_{it})}^{\text{carried into } t}}_{\text{net savings}} = \overbrace{R_{it}(f_{it})}^{\text{“consumable” income}}$$

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low}, \text{Middle}, \text{High}\}$
  - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}\,(i \text{ at risk of friction at } t)$, $f_{it} \equiv \mathbf{1}\,(i \text{ subject to friction at } t)$
  - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
  - f_{it} may depend on past choices (e.g. for-profit college attendance)
- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \text{Wealth}_{it+1}(f_{it}) - \text{Wealth}_{it}(f_{it}) = R_{it}(f_{it})$$

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) \qquad - R_{it}(f_{it})$$

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low}, \text{Middle}, \text{High}\}$
  - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income

- $s_{it} \equiv \mathbf{1}\,(i \text{ at risk of friction at } t)$, $f_{it} \equiv \mathbf{1}\,(i \text{ subject to friction at } t)$
  - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
  - f_{it} may depend on past choices (e.g. for-profit college attendance)

- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \text{Wealth}_{it+1}(f_{it}) - \text{Wealth}_{it}(f_{it}) = R_{it}(f_{it})$$

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\,A_{it}(0) - R_{it}(f_{it})\,A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\,A_{it}(0) - R_{it}(f_{it})\,A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\, A_{it}(0) - R_{it}(f_{it})\, A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\, A_{it}(0) - R_{it}(f_{it})\, A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Attending 2-Year For-Profit Colleges: Reduces Labor Income

Measuring Financial Impact of Frictions

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\,A_{it}(0) - R_{it}(f_{it})\,A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Not Investing in Stock Mutual Funds: Reduces Capital Income

Measuring Financial Impact of Frictions

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\,A_{it}(0) - R_{it}(f_{it})\,A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Not Claiming EITC Benefits: Reduces Benefits

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\, A_{it}(0) - R_{it}(f_{it})\, A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Paying Credit Card Late Fees: Creates direct costs from additional fees

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\,A_{it}(0) - R_{it}(f_{it})\,A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Not Quitting Smoking: Creates direct costs from cigarettes and increases mortality

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0)\, A_{it}(0) - R_{it}(f_{it})\, A_{it}(f_{it}), \qquad A_{it}(f_{it}) = \mathbf{1}\,(i \text{ alive at } t)$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels
- **Challenge**: fully accounting for interaction between channels + dynamic effects
  - Ignore dynamics in this analysis, and incorporate in **life cycle model**

Key Parameters that We Estimate for Each Friction

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}\left(s_{it} = 1 \mid A_{it}(f_{it}) = 1,\, j_{it} = j\right)$

Key Parameters that We Estimate for Each Friction

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}\left(s_{it} = 1 \mid A_{it}(f_{it}) = 1,\, j_{it} = j\right)$

Friction Prevalence: $F_j \equiv \mathbb{P}\left(f_{it} = 1 \mid s_{it} = 1,\, A_{it}(f_{it}) = 1,\, j_{it} = j\right)$

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}\left(s_{it} = 1 \mid A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Friction Prevalence: $F_j \equiv \mathbb{P}\left(f_{it} = 1 \mid s_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Average Loss: $L_j \equiv \dfrac{\mathbb{E}\left[f_{it} l_{it}(f_{it}) \mid j_{it} = j\right]}{\mathbb{E}\left[A_{it}(f_{it}) f_{it} \mid j_{it} = j\right]}$

Key Parameters that We Estimate for Each Friction

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}\left(s_{it} = 1 \mid A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Friction Prevalence: $F_j \equiv \mathbb{P}\left(f_{it} = 1 \mid s_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Average Loss: $L_j \equiv \dfrac{\mathbb{E}\left[f_{it} l_{it}(f_{it}) \mid j_{it} = j\right]}{\mathbb{E}\left[A_{it}(f_{it}) f_{it} \mid j_{it} = j\right]}$

- Mortality **not** affected $\Rightarrow$ average loss simplifies to

$$L_j = \mathbb{E}\left[R_{it}(0) - R_{it}(f_{it}) \mid f_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right]$$

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}\left(s_{it} = 1 \mid A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Friction Prevalence: $F_j \equiv \mathbb{P}\left(f_{it} = 1 \mid s_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Average Loss: $L_j \equiv \dfrac{\mathbb{E}\left[f_{it} l_{it}(f_{it}) \mid j_{it} = j\right]}{\mathbb{E}\left[A_{it}(f_{it}) f_{it} \mid j_{it} = j\right]}$

- Mortality **not** affected $\Rightarrow$ average loss simplifies to

$$L_j = \mathbb{E}\left[R_{it}(0) - R_{it}(f_{it}) \mid f_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right]$$

- Mortality **is** affected $\Rightarrow$ who's alive changes, and average loss is

$$L_j = \frac{\text{total financial impact among } j}{\text{\# of affected observations in } j \textit{ that are currently alive}}$$

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}\left(s_{it} = 1 \mid A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Friction Prevalence: $F_j \equiv \mathbb{P}\left(f_{it} = 1 \mid s_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right)$

Average Loss: $L_j \equiv \dfrac{\mathbb{E}\left[f_{it} l_{it}(f_{it}) \mid j_{it} = j\right]}{\mathbb{E}\left[A_{it}(f_{it}) f_{it} \mid j_{it} = j\right]}$

Unconditional Loss: $\Delta_j \equiv \dfrac{\mathbb{E}\left[f_{it} l_{it}(f_{it}) \mid j_{it} = j\right]}{\mathbb{E}\left[A_{it}(f_{it}) \mid j_{it} = j\right]} = S_j F_j L_j$

- Mortality **not** affected $\Rightarrow$ average loss simplifies to

$$L_j = \mathbb{E}\left[R_{it}(0) - R_{it}(f_{it}) \mid f_{it} = 1,\ A_{it}(f_{it}) = 1,\ j_{it} = j\right]$$

- Mortality **is** affected $\Rightarrow$ who's alive changes, and average loss is

$$L_j = \frac{\text{total financial impact among } j}{\text{\# of affected observations in } j \textit{ that are currently alive}}$$

Additional Parameters and Caveats

- Important parameters in life cycle model, but not needed for empirical approach:
  - **Correlation** of different frictions within-individual (in progress)
  - **Autocorrelation** of frictions within individual (for now: permanent)
  - **Sophistication** about frictions (for now: full sophistication)
- Caveats to both of our approaches:
  - Do not capture spillovers, GE effects, or **cross-subsidies** (could potentially incorporate)
  - Removing a particular (set of) friction(s) may not be **feasible** given microfoundations

EXAMPLES

Example: Paying Late Fees

- s_{it} = having a credit card, f_{it} = paying at least one late fee
  - Shares S_j and F_j measured by income group (CFPB Making Ends Meet)
  - Adjust for survey underreporting of credit cards (Karlan-Zinman 2008)
- Financial impact comes from λ_{it} = direct losses from paying late fees
  - No micro data on fees $\Rightarrow$ estimate one L_j for all j (CFPB CCM Report)
  - L_j = (average late fee) $\times$ (# fees per card $|$ fees > 1) $\times$ (# cards per cardholder)

- s_{it} = having a credit card, f_{it} = paying at least one late fee
  - Shares S_j and F_j measured by income group (CFPB Making Ends Meet)
  - Adjust for survey underreporting of credit cards (Karlan-Zinman 2008)
- Financial impact comes from λ_{it} = direct losses from paying late fees
  - No micro data on fees $\Rightarrow$ estimate one L_j for all j (CFPB CCM Report)
  - L_j = (average late fee) $\times$ (# fees per card | fees $>$ 1) $\times$ (# cards per cardholder)

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.463	0.311	\$406	\$57
	(0.013)	(0.017)	[381, 425]	[49, 64]
Middle Income	0.736	0.251	\$406	\$73
	(0.004)	(0.010)	[381, 425]	[70, 78]
High Income	0.827	0.133	\$406	\$44
	(0.007)	(0.011)	[381, 425]	[38, 50]

Example: Attending 2-Year For-Profit Colleges

- s_{it} = attending college, f_{it} = attending 2-year for-profit
  - Shares S_j and F_j measured by parental income group (BPS 12/17, Snyder et al. 2019)
  - Translate to child income group using intergenerational correlation (Haider-Solon 2006)
- Financial impact combines existing literature (Cellini-Turner 2019) + simulation
  - 1.5pp increase in unemployment probability $\Rightarrow \downarrow$ Labor Income
  - 11.3% earnings loss conditional on employment $\Rightarrow \downarrow$ Labor Income
  - $6,100 increase in student debt (repaid under default plan) $\Rightarrow \lambda_{it}$

Example: Attending 2-Year For-Profit Colleges

- s_{it} = attending college, f_{it} = attending 2-year for-profit
  - Shares S_j and F_j measured by parental income group (BPS 12/17, Snyder et al. 2019)
  - Translate to child income group using intergenerational correlation (Haider-Solon 2006)
- Financial impact combines existing literature (Cellini-Turner 2019) + simulation
  - 1.5pp increase in unemployment probability $\Rightarrow \downarrow$ Labor Income
  - 11.3% earnings loss conditional on employment $\Rightarrow \downarrow$ Labor Income
  - \$6,100 increase in student debt (repaid under default plan) $\Rightarrow \lambda_{it}$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.263	0.236	\$1,364	\$85
	(0.016)	(0.014)	[1,116, 1,613]	[64, 106]
Middle Income	0.277	0.094	\$2,755	\$72
	(0.010)	(0.007)	[1,961, 3,550]	[47, 97]
High Income	0.138	0.037	\$1,781	\$9
	(0.012)	(0.011)	[447, 3,116]	[-14, 32]

Example: Not Investing in Stock Mutual Funds

- s_{it} = have financial wealth $\geq W_{-} \equiv \min\{2 \text{ mo. salary}, \$2K\}$ (SCF, Sabat-Gallagher 2019)
- f_{it} = not investing any wealth in diversified stocks (SCF)
  - Unlikely to be reflective of non-standard preferences (Choukhmane-de Silva 2026)
- Financial impact comes from $\downarrow$ Capital Income
  - $\ell_{it}(f_{it}) =$ counterfactual equity share $\times$ (financial wealth $- W_{-}$)
  - Counterfactual equity share = average age-specific equity share of those with $f_{it} = 0$

Example: Not Investing in Stock Mutual Funds

- s_{it} = have financial wealth $\geq W_{-} \equiv \min\{2 \text{ mo. salary}, \$2K\}$ (SCF, Sabat-Gallagher 2019)
- f_{it} = not investing any wealth in diversified stocks (SCF)
  - Unlikely to be reflective of non-standard preferences (Choukhmane-de Silva 2026)
- Financial impact comes from $\downarrow$ Capital Income
  - $\ell_{it}(f_{it}) =$ counterfactual equity share $\times$ (financial wealth $- W_{-}$)
  - Counterfactual equity share = average age-specific equity share of those with $f_{it} = 0$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.326	0.646	\$354	\$75
	(0.008)	(0.015)	[297, 436]	[61, 92]
Middle Income	0.643	0.305	\$959	\$188
	(0.006)	(0.008)	[887, 1,039]	[171, 206]
High Income	0.867	0.129	\$2,615	\$293
	(0.006)	(0.007)	[2,193, 3,076]	[239, 354]

- s_{it} = smoke before 50 and want to quit, f_{it} = want to quit before 50 but don't (NHIS)
- Choose age 50 as critical year for two reasons:
  1. Those who successfully quit before 50 have similar mortality to never-smokers
  2. Those who quit after 50 have similar mortality to non-smokers
- Financial impact comes from mortality and direct losses
  - Simulate increased (∼2x) mortality prob. by income–age $\Rightarrow A_{it}(0) - A_{it}(f_{it}) \geq 0$
  - Cost of cigarettes ∼ \$1.2K/year $\Rightarrow \lambda_{it}$

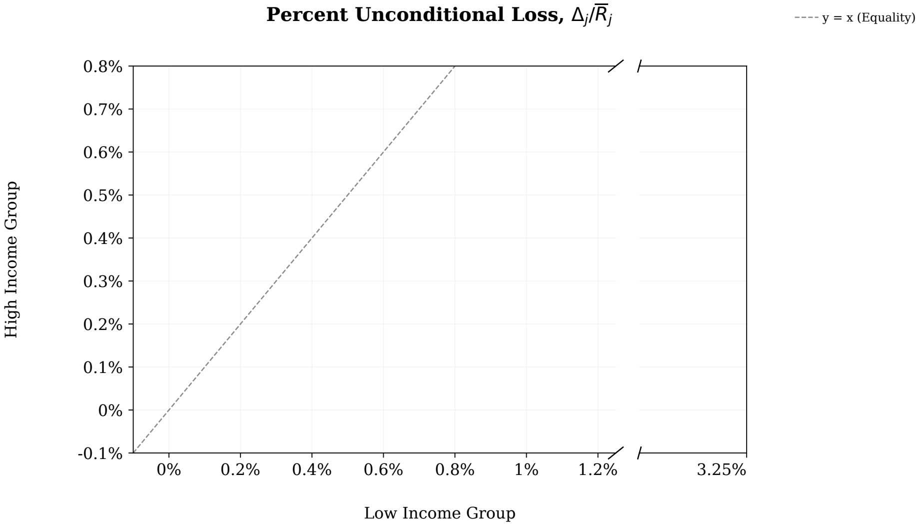
- s_{it} = smoke before 50 and want to quit, f_{it} = want to quit before 50 but don't (NHIS)
- Choose age 50 as critical year for two reasons:
  1. Those who successfully quit before 50 have similar mortality to never-smokers
  2. Those who quit after 50 have similar mortality to non-smokers
- Financial impact comes from mortality and direct losses
  - Simulate increased ($\sim$2x) mortality prob. by income–age $\Rightarrow A_{it}(0) - A_{it}(f_{it}) \geq 0$
  - Cost of cigarettes $\sim$ \$1.2K/year $\Rightarrow \lambda_{it}$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.414	0.563	\$2,853	\$665
	(0.027)	(0.031)	[2,636, 3,071]	[530, 801]
Middle Income	0.350	0.414	\$3,599	\$521
	(0.020)	(0.028)	[3,190, 4,008]	[396, 647]
High Income	0.298	0.240	\$5,211	\$373
	(0.024)	(0.047)	[2,342, 8,081]	[144, 602]

Outline

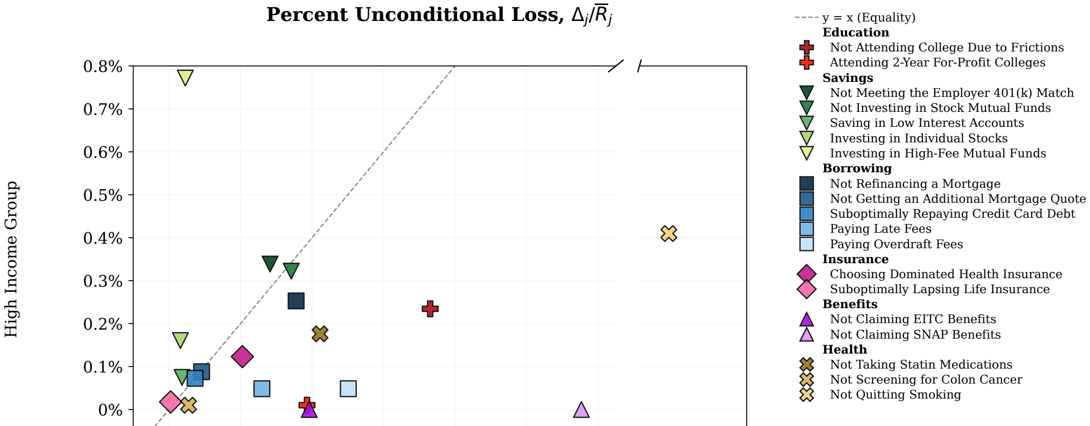
1. Empirical Framework and Examples (Approach #1)
2. Results (Approach #1)
3. Life Cycle Model (Sketch) and Results (Approach #2)
4. Conclusion

Financial Impacts as % of Permanent Consumable Income



▸ Dollars

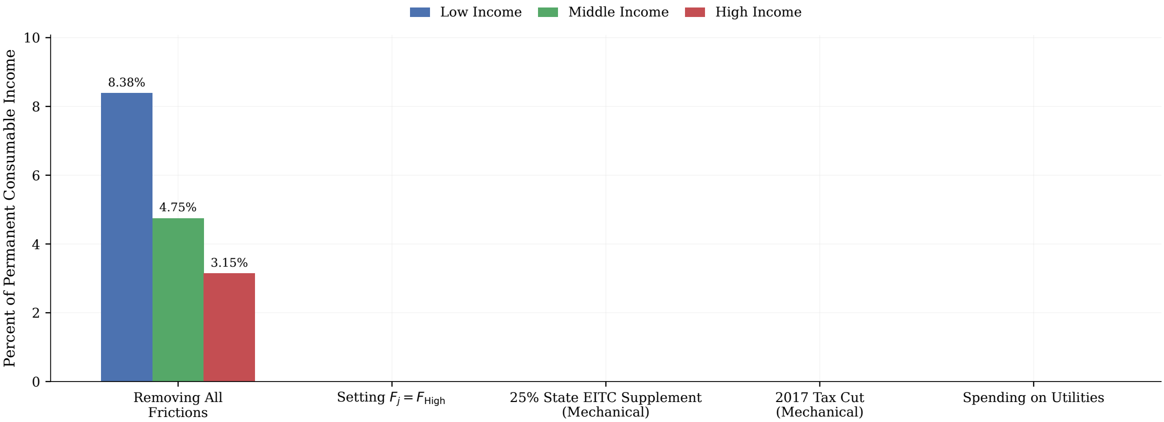
Financial Impacts as % of Permanent Consumable Income



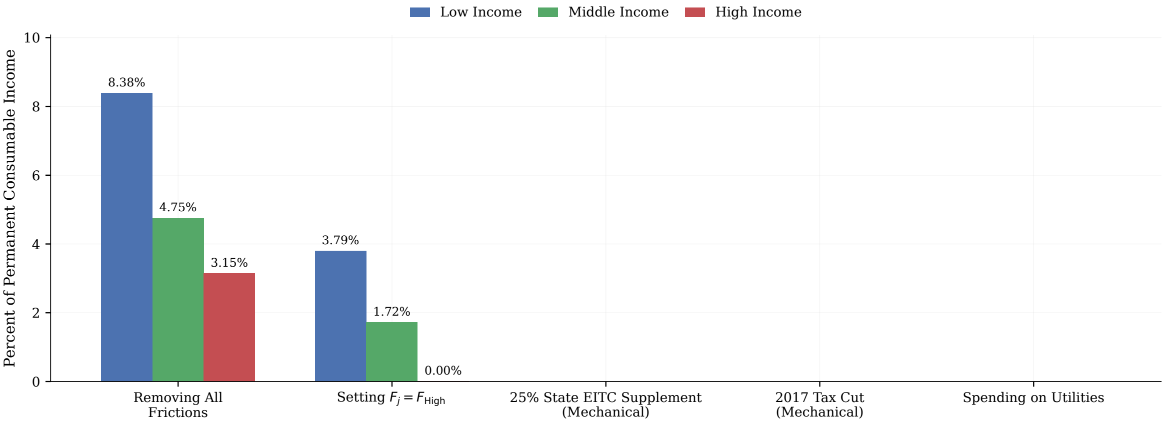
Income Group	Average	All
Low	0.44%	8.38%
High	0.17%	3.15%

Dollars

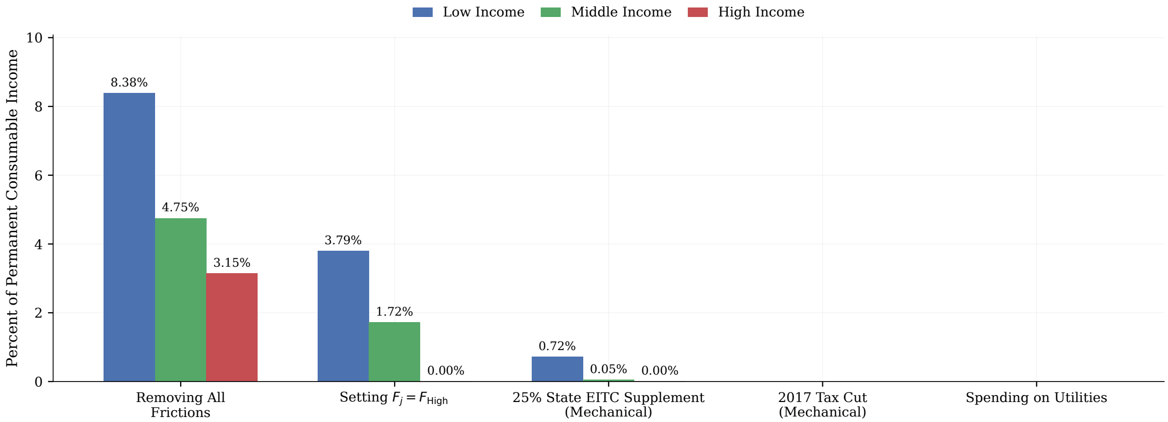
Benchmarking Total Losses from Frictions



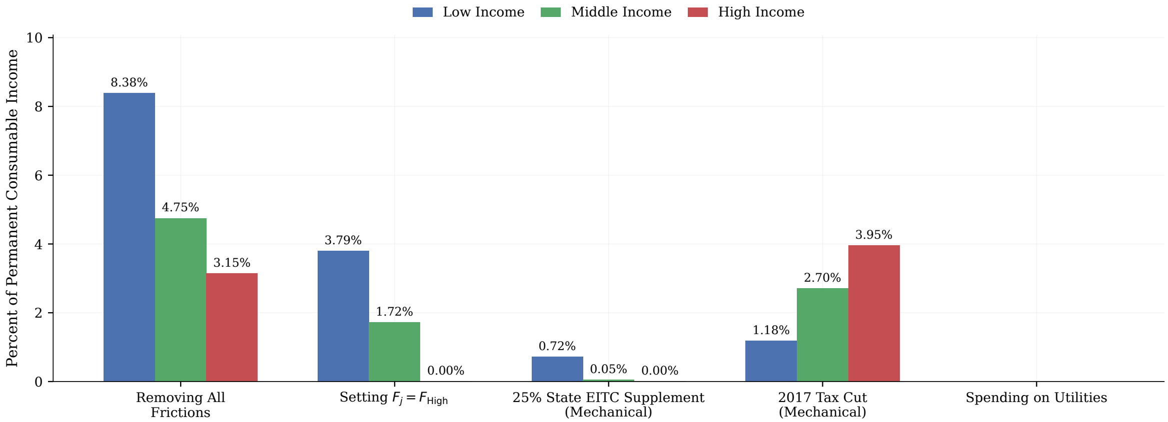
Benchmarking Total Losses from Frictions

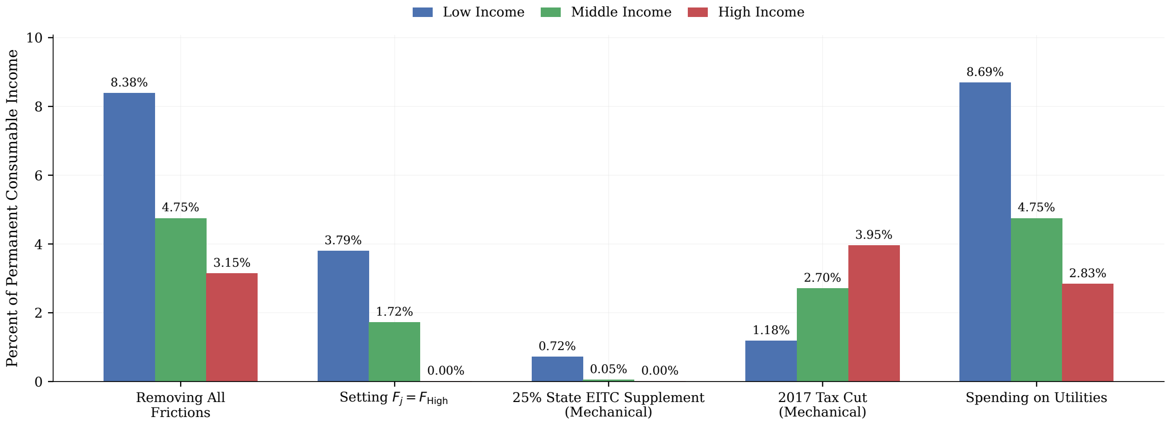


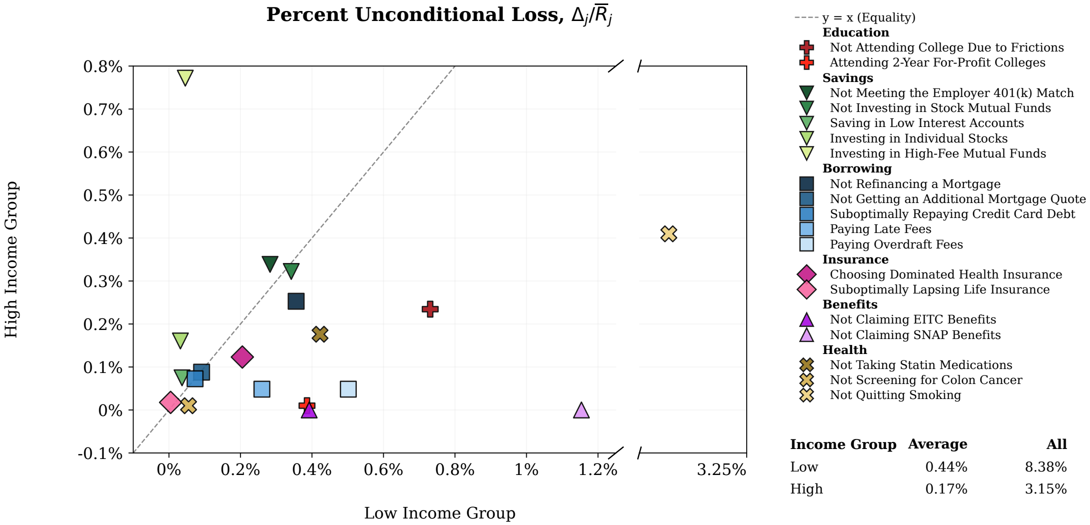
Benchmarking Total Losses from Frictions



Benchmarking Total Losses from Frictions



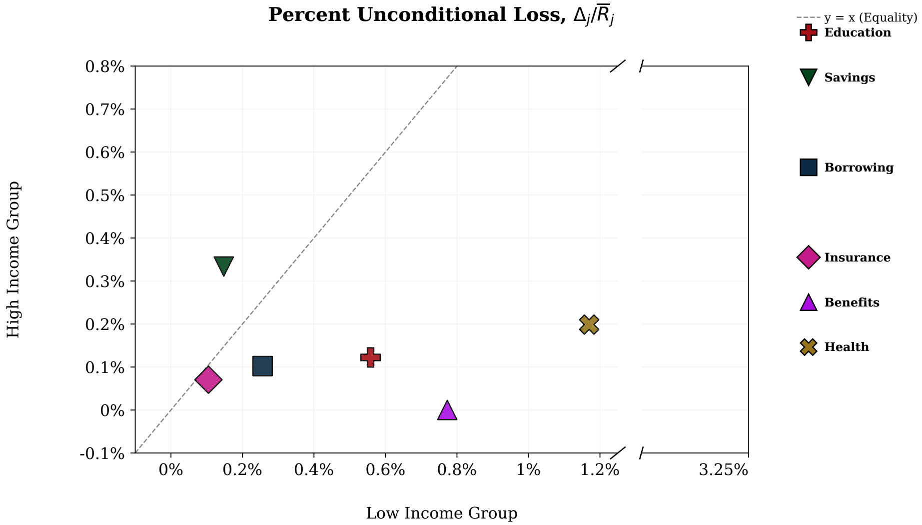

Low Income
Middle Income
High Income
Percent of Permanent Consumable Income
0
2
4
6
8
10
8.38%
4.75%
3.15%
3.79%
1.72%
0.00%
0.72%
0.05%
0.00%
1.18%
2.70%
3.95%
8.69%
4.75%
2.83%
Removing All Frictions
Setting $F_j = F_{\text{High}}$
25% State EITC Supplement (Mechanical)
2017 Tax Cut (Mechanical)
Spending on Utilities



Income Group	Average	All
Low	0.44%	8.38%
High	0.17%	3.15%

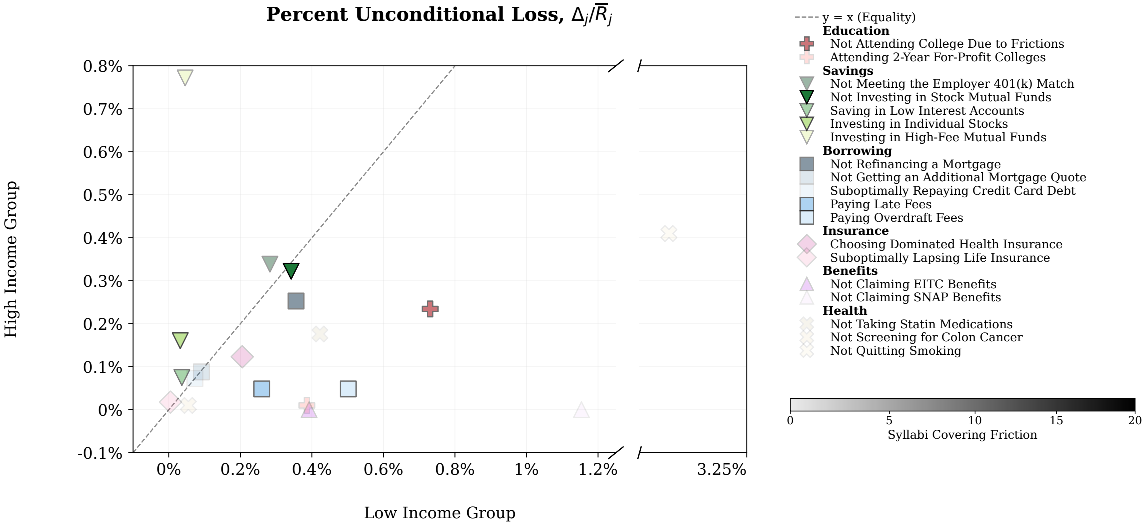
▸ Dollars

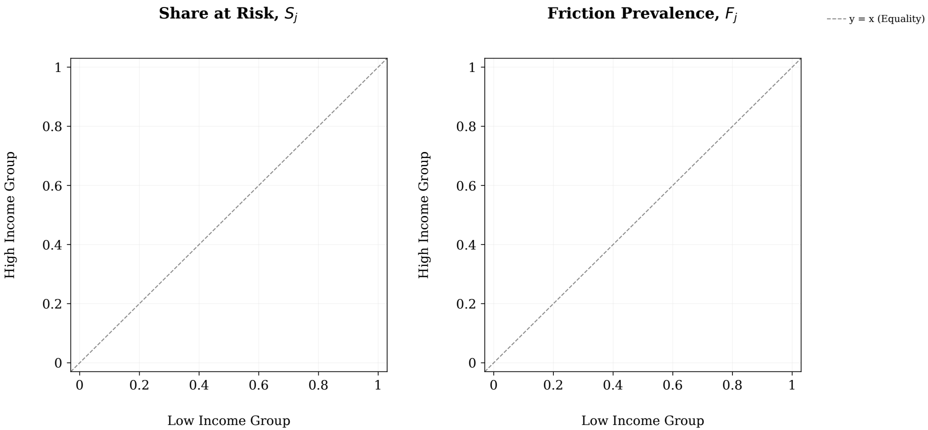
Financial Impacts: Category Averages



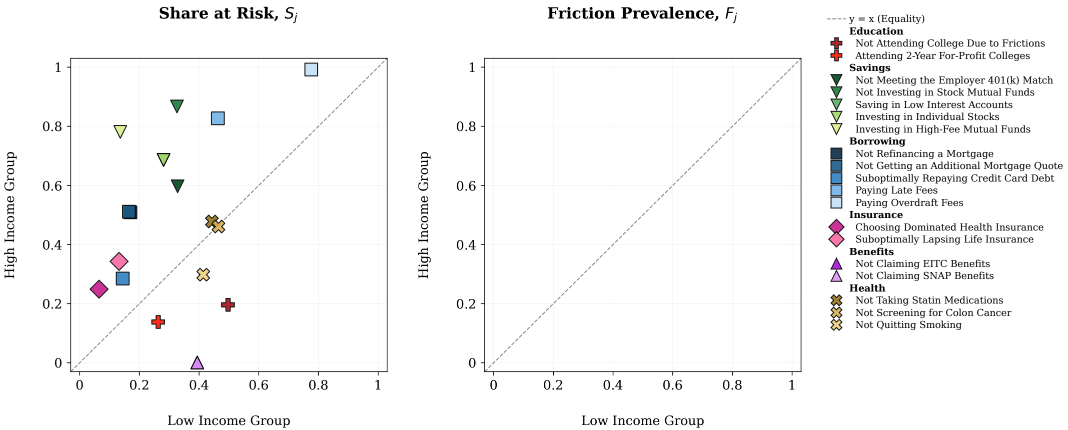
▸ Dollars

Financial Impact of Frictions: Syllabus Coverage

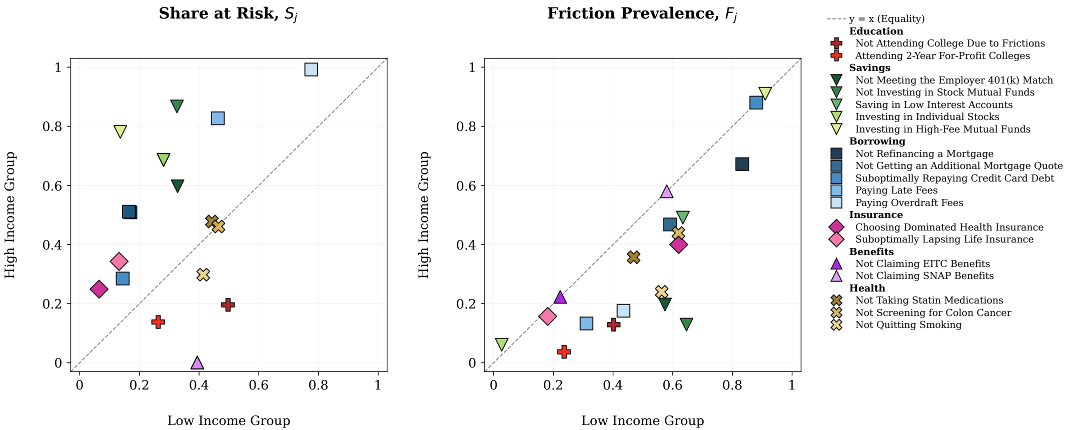



Share at Risk, S_j
Friction Prevalence, F_j
y = x (Equality)
High Income Group
Low Income Group
0
0.2
0.4
0.6
0.8
1

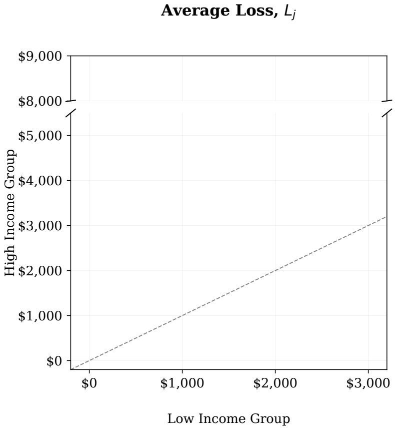
Decomposing Financial Impact: Shares and Prevalence

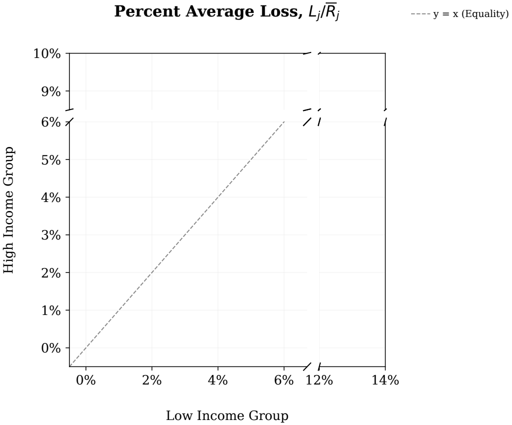


Decomposing Financial Impact: Shares and Prevalence

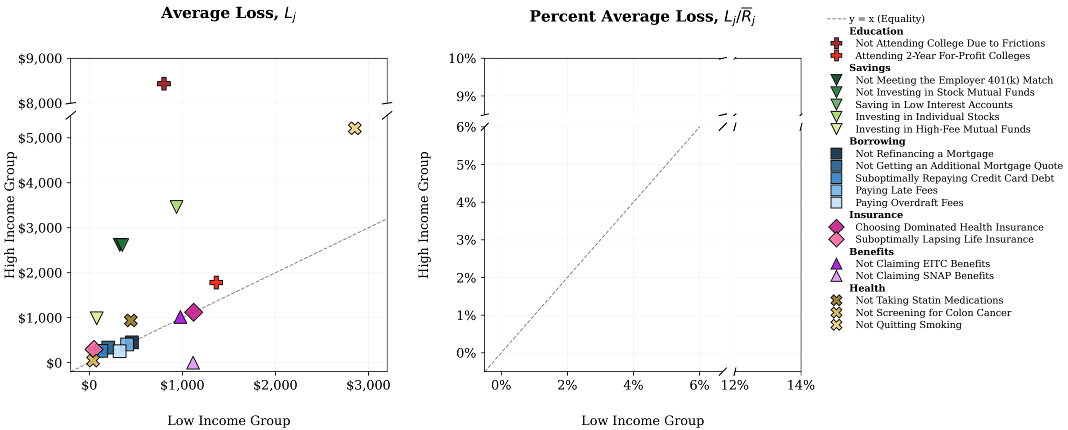


Decomposing Financial Impact: Average Losses

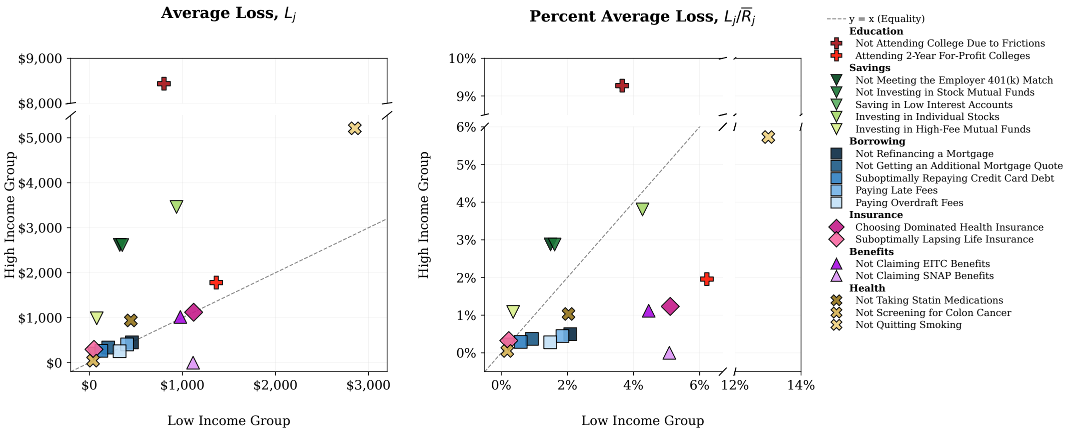


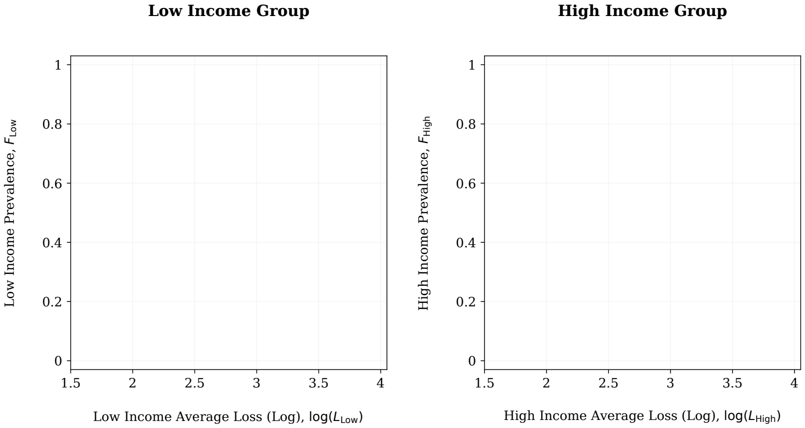


Decomposing Financial Impact: Average Losses

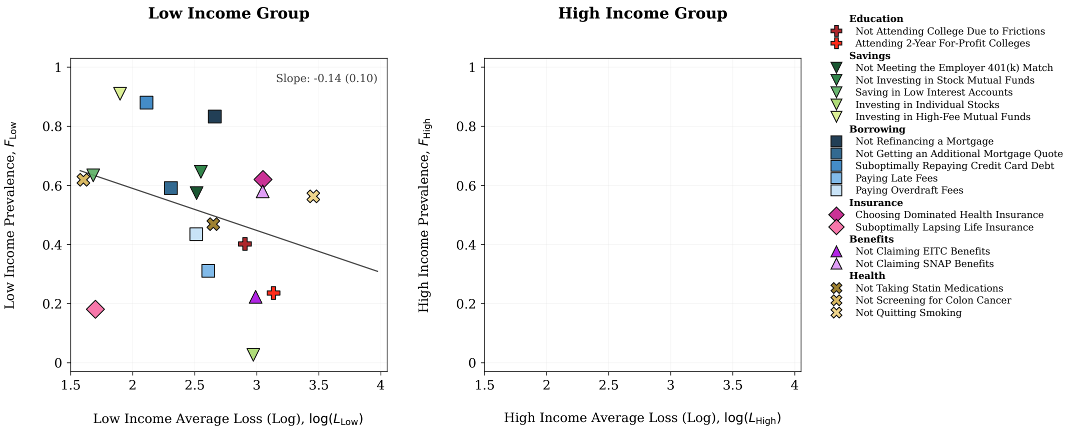


Decomposing Financial Impact: Average Losses

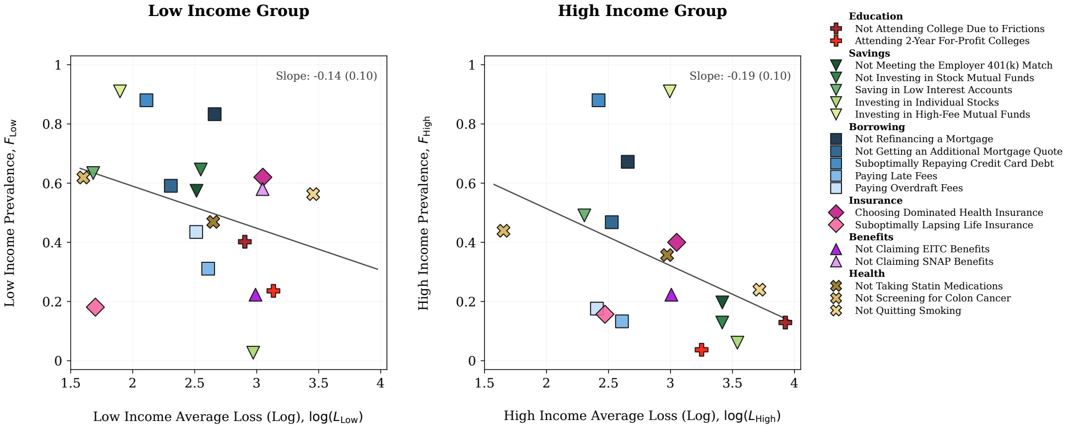



Low Income Group
High Income Group
Low Income Prevalence, F_{Low}
High Income Prevalence, F_{High}
1
0.8
0.6
0.4
0.2
0
1.5
2
2.5
3
3.5
4
Low Income Average Loss (Log), $\log(L_{\text{Low}})$
High Income Average Loss (Log), $\log(L_{\text{High}})$

Relationship Between Prevalence and Average Loss



Relationship Between Prevalence and Average Loss



Results In Progress

1. **Evolution** of key parameters over time
   - Losses seem to respond to policies (e.g. for-profit certifications, public health efforts)
2. **Correlation** of prevalence across-frictions within-individual
   - Currently have estimated 6 bivariate correlations with mean 0.032 (SE 0.006)
3. Comparison of losses with **forecasts** from a survey of academics
4. Comparison of losses with coverage in **academic journals** and **media**

Results In Progress

1. **Evolution** of key parameters over time
   - Losses seem to respond to policies (e.g. for-profit certifications, public health efforts)
2. **Correlation** of prevalence across-frictions within-individual
   - Currently have estimated 6 bivariate correlations with mean 0.032 (SE 0.006)
3. Comparison of losses with **forecasts** from a survey of academics
4. Comparison of losses with coverage in **academic journals** and **media**

... More ideas welcome!

Outline

1. Empirical Framework and Examples (Approach #1)
2. Results (Approach #1)
3. Life Cycle Model (Sketch) and Results (Approach #2)
4. Conclusion

Model Summary

- Reasons to develop a life cycle model:
  1. Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
  2. Removing a friction likely creates **endogenous** responses
  3. Study affects on aggregate **consumption** and **wealth** inequality

MODEL SUMMARY

- Reasons to develop a life cycle model:
  1. Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
  2. Removing a friction likely creates **endogenous** responses
  3. Study affects on aggregate **consumption** and **wealth** inequality

- Key model ingredients (partial equilibrium):
  - Individuals ex-ante heterogeneous in permanent income and education
  - Individuals make dynamic consumption-savings and portfolio choice decisions
  - Available assets: liquid savings, 401(k), credit cards, housing
  - Non-Gaussian income risk, mortality risk, realistic social insurance system

Model Summary

- Reasons to develop a life cycle model:
  1. Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
  2. Removing a friction likely creates **endogenous** responses
  3. Study affects on aggregate **consumption** and **wealth** inequality

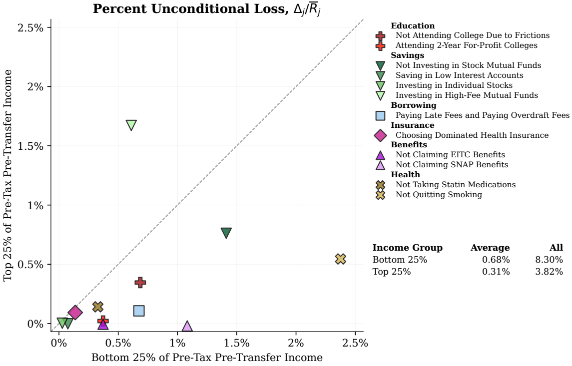
- Key model ingredients (partial equilibrium):
  - Individuals ex-ante heterogeneous in permanent income and education
  - Individuals make dynamic consumption-savings and portfolio choice decisions
  - Available assets: liquid savings, 401(k), credit cards, housing
  - Non-Gaussian income risk, mortality risk, realistic social insurance system

- Individuals subject to frictions $\mathbf{f}_i \in \{0, 1\}^{19}$ based on **empirical prevalence**
  - Permanent throughout life, full sophistication, no correlation (for now)

Model Summary

- Reasons to develop a life cycle model:
  1. Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
  2. Removing a friction likely creates **endogenous** responses
  3. Study affects on aggregate **consumption** and **wealth** inequality
- Key model ingredients (partial equilibrium):
  - Individuals ex-ante heterogeneous in permanent income and education
  - Individuals make dynamic consumption-savings and portfolio choice decisions
  - Available assets: liquid savings, 401(k), credit cards, housing
  - Non-Gaussian income risk, mortality risk, realistic social insurance system
- Individuals subject to frictions $\mathbf{f}_i \in \{0, 1\}^{19}$ based on **empirical prevalence**
  - Permanent throughout life, full sophistication, no correlation (for now)
- **Key counterfactuals**: Set $\mathbf{f}_i(n) = 0$ for each friction n and compare with data

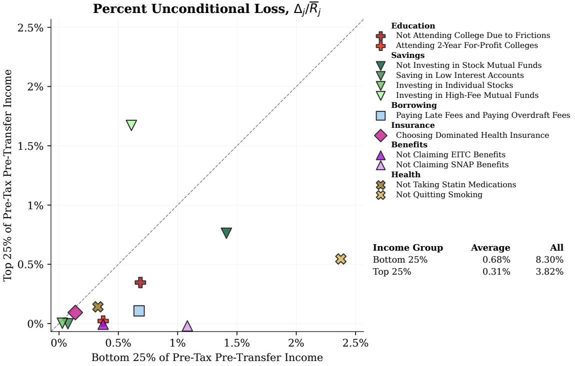
Life Cycle Model



Income Group	Average	All
Bottom 25%	0.68%	8.30%
Top 25%	0.31%	3.82%

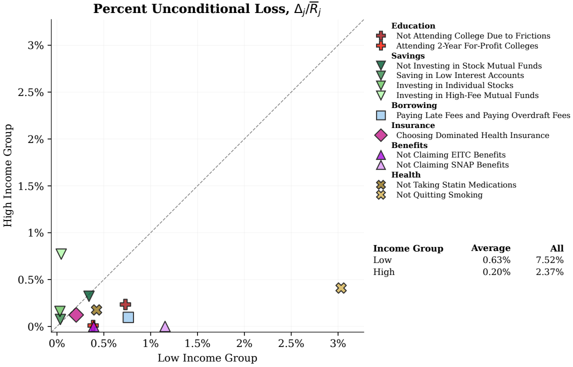
Financial Impact of Frictions: Two Approaches

Life Cycle Model



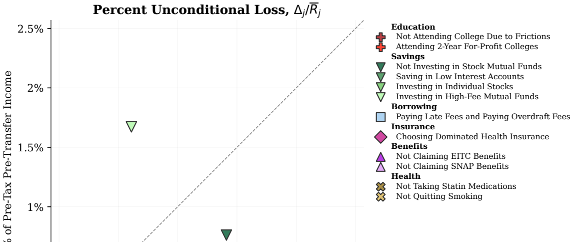
Income Group	Average	All
Bottom 25%	0.68%	8.30%
Top 25%	0.31%	3.82%

Empirical Estimates



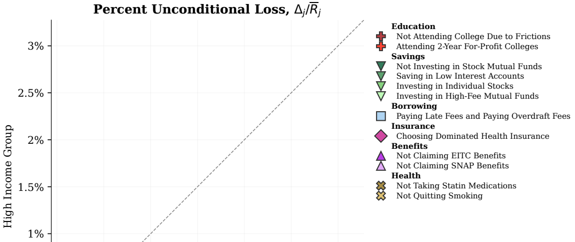
Income Group	Average	All
Low	0.63%	7.52%
High	0.20%	2.37%

Life Cycle Model



Income Group	Average	All
Bottom 25%	0.68%	8.30%
Top 25%	0.31%	3.82%

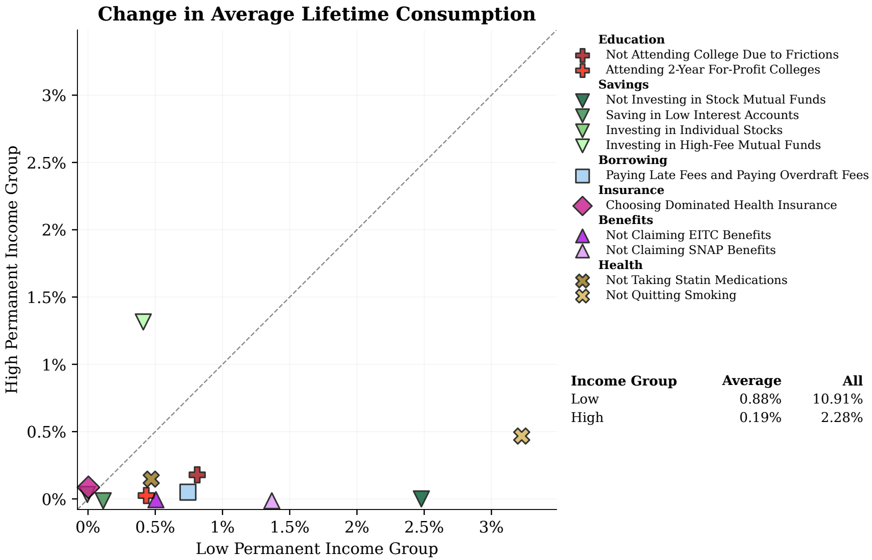
Empirical Estimates



Income Group	Average	All
Low	0.63%	7.52%
High	0.20%	2.37%

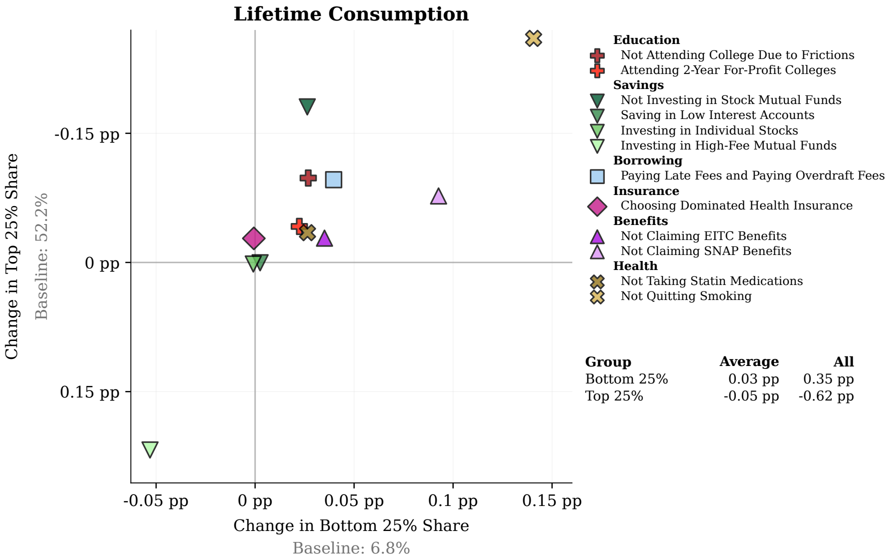
Losses are much larger for **savings** frictions, due to **dynamic** effects

Frictions Have Large Effects on Lifetime Consumption



Income Group	Average	All
Low	0.88%	10.91%
High	0.19%	2.28%

Frictions Reduce Lifetime Consumption Inequality



Group	Average	All
Bottom 25%	0.03 pp	0.35 pp
Top 25%	-0.05 pp	-0.62 pp

Outline

1. Empirical Framework and Examples (Approach #1)
2. Results (Approach #1)
3. Life Cycle Model (Sketch) and Results (Approach #2)
4. Conclusion

1. Decision-making frictions create meaningful losses in aggregate
   - Aggregate losses are **8%** (**3%**) for low (high) income = expenditure on utilities
   - Largest (as of now): smoking, fund fees, college attendance, SNAP benefits
2. Losses from decision-making frictions are **regressive**
   - High-income have more to lose (L), but low-income **prevalence** (F) is higher
   - Main exception: Savings frictions, which are most covered by syllabi
   - **Model**: Removing frictions decreases inequality in lifetime consumption

Summary

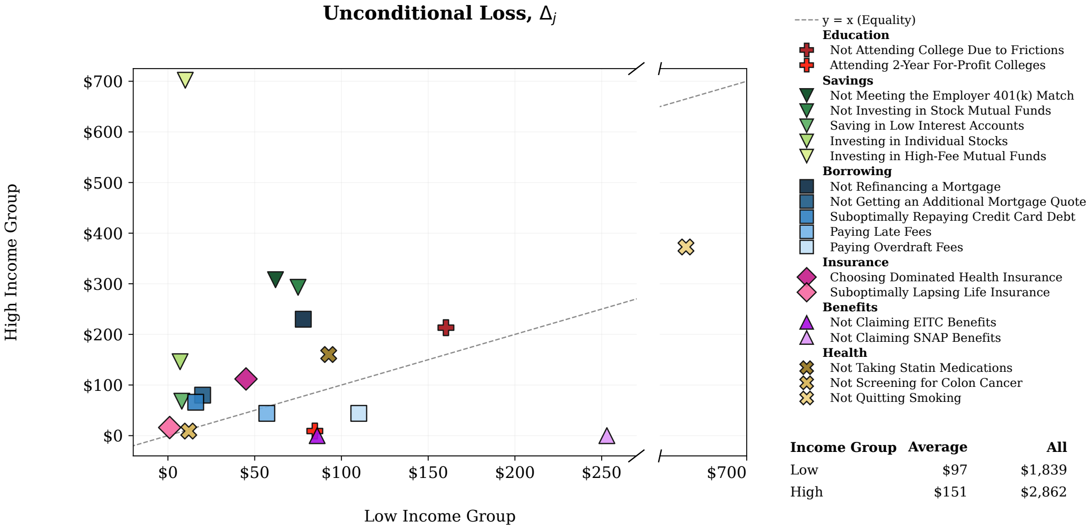
1. Decision-making frictions create meaningful losses in aggregate
   - Aggregate losses are **8%** (**3%**) for low (high) income = expenditure on utilities
   - Largest (as of now): smoking, fund fees, college attendance, SNAP benefits

2. Losses from decision-making frictions are **regressive**
   - High-income have more to lose (L), but low-income **prevalence** (F) is higher
   - Main exception: Savings frictions, which are most covered by syllabi
   - **Model**: Removing frictions decreases inequality in lifetime consumption

... Much more to come!

Thank You!

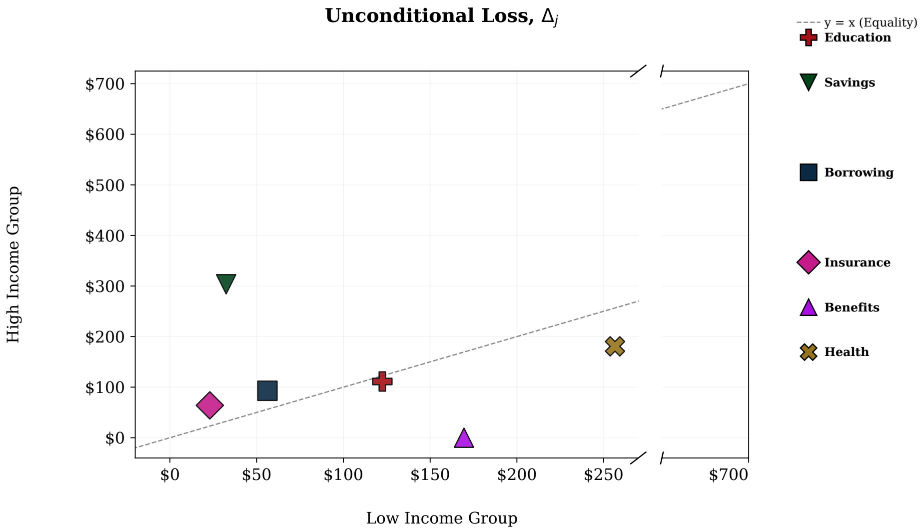
sdellavi@berkeley.edu
tdesilva@stanford.edu
rohanjha@g.harvard.edu



Income Group	Average	All
Low	$97	$1,839
High	$151	$2,862

Back

Financial Impacts: Category Averages in Dollars



Back